Karthik P

Pathanamthitta, Kerala, India

7012041942 itskp2004@gmail.com Linkedln Github Portfolio

TECHNICAL SKILLS

- **Languages**: Java, Python, C, C++, HTML/CSS, JavaScript
- **Data Structures & Algorithms**: Arrays, Trees, Heaps, Graphs, Hashing
- **Technologies/Frameworks/Libraries**: Node.js, React.js, Electron.js, Spring Boot, Docker, PostgreSQL, Next.js, MySQL, Tablue

EDUCATION

Amrita Vishwa Vidyapeetham University

*B.Tech Computer Science & Engineering (CGPA of **7.13**)* **2023-2027**

St. Mary's Residential School, Pathanapuram, Kollam

*Higher Secondary School, ISC, Science with Computer Science (**86%**)* **2022-2023**

*Secondary School, ICSE (**96.2%**)*

2020-2021

PROJECTS

MARINA | *Java, Python* | Group Project

- Designed and implemented an intelligent vehicular edge computing system using LSTM-based mobility prediction models to anticipate vehicle movement patterns and enable proactive task scheduling decisions.
- Reduced overall task execution cost while meeting strict deadline constraints by dynamically allocating computational workloads between mobile vehicles and roadside base stations based on predicted mobility and resource availability.

LeadGraph: Social-to-Lead AI Agent | *Python, LangGraph, Google Gemini, FAISS* | Individual Project

- Built a stateful conversational workflow using LangGraph to manage dialogue loops, featuring an LLM-driven intent classification engine that seamlessly transitions between natural conversation and structured data extraction.
- Implemented a local FAISS-powered Retrieval-Augmented Generation (RAG) pipeline to autonomously answer product queries, executing simulated lead capture tool calls while preserving cross-turn context memory.

Secure Prescription | *React, Spring Boot, PostgreSQL, Docker* | Individual Project

- Engineered a full-stack secure prescription system with a high-integrity architecture, implementing Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC) to ensure unauthorized data tampering is prevented across all user workflows.
- Designed a minimalist, responsive React UI featuring streamlined dashboards and containerized the entire stack (Spring Boot, PostgreSQL, React) using Docker Compose to ensure seamless, consistent deployment across different environments.

Hybrid Stock Predictor Web App | *React, FastAPI, FinBERT, yfinance* | Individual Project

- Developed a full-stack prediction platform featuring real-time sentiment analysis of financial news using FinBERT, with dynamic weighting logic that prioritizes news signals (70%) over technical indicators when recent news exists.
- Integrated live stock data from yfinance and user watchlist management with FastAPI backend, delivering daily BUY/SELL/HOLD signals through an interactive React dashboard.

Impact of Weather on Retail Sales | *Python, Pandas, Statsmodels, Chart.js* | Group Project

- Integrated 2014–2018 grocery sales data with real-world weather records to quantify how temperature, rainfall, and seasonality influence consumer behavior, applying EDA, hypothesis testing, and OLS regression for actionable insights.
- Developed high-resolution visualizations and an interactive Chart.js dashboard to present key findings and support strategic business decisions.

CERTIFICATIONS

- IBM AI Engineering Professional Certificate (13-course series covering ML, Deep Learning, Generative AI, LLMs, and RAG systems)